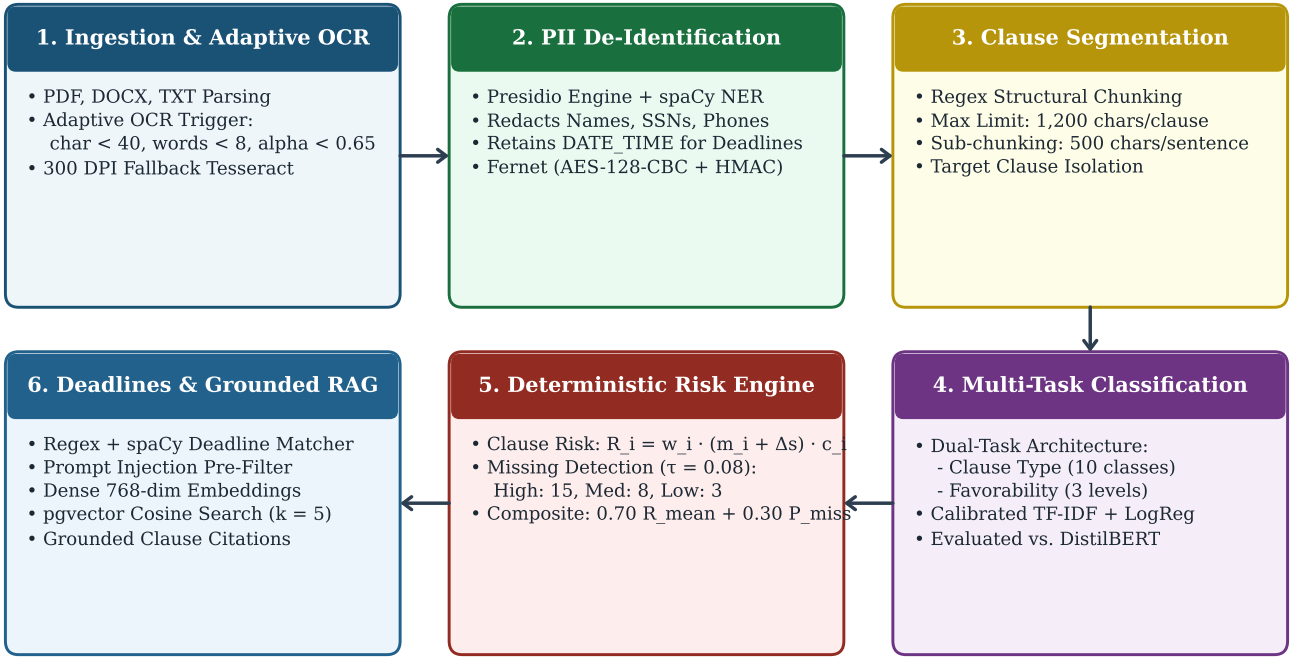
1. Ingestion & Adaptive OCR
• PDF, DOCX, TXT Parsing
• Adaptive OCR Trigger:
char < 40, words < 8, alpha < 0.65
• 300 DPI Fallback Tesseract
2. PII De-Identification
• Presidio Engine + spaCy NER
• Redacts Names, SSNs, Phones
• Retains DATE_TIME for Deadlines
• Fernet (AES-128-CBC + HMAC)
3. Clause Segmentation
• Regex Structural Chunking
• Max Limit: 1,200 chars/clause
• Sub-chunking: 500 chars/sentence
• Target Clause Isolation
4. Multi-Task Classification
• Dual-Task Architecture:
- Clause Type (10 classes)
- Favorability (3 levels)
• Calibrated TF-IDF + LogReg
• Evaluated vs. DistilBERT
5. Deterministic Risk Engine
• Clause Risk: R_i = w_i · (m_i + Δs) · c_i
• Missing Detection (τ = 0.08):
High: 15, Med: 8, Low: 3
• Composite: 0.70 R_mean + 0.30 P_miss
6. Deadlines & Grounded RAG
• Regex + spaCy Deadline Matcher
• Prompt Injection Pre-Filter
• Dense 768-dim Embeddings
• pgvector Cosine Search (k = 5)
• Grounded Clause Citations